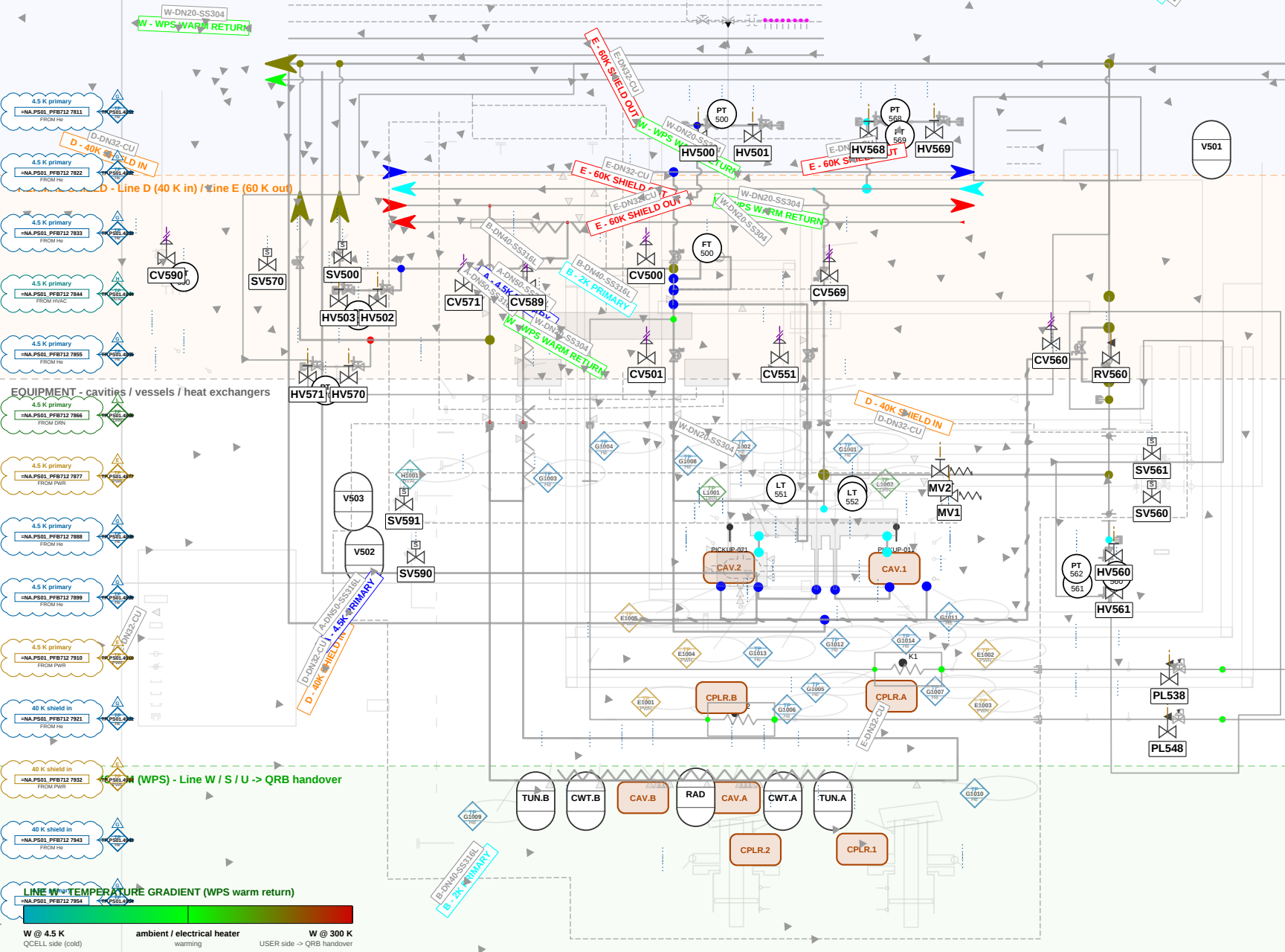


P&ID - QCELL / LB - Cryogenic Circuits (40 K / 4.5 K / 2 K + HX)

CONTROL-CENTRIC

COLD HEADER - Line A (4.5 K) / Line B (2 K)



INTERACTIVE LEGEND

toggle: layer "36_Legend_INTERACTIVE"

LINE SPECIFICATION TABLE

LINE	TEMP	PRESS	DN	MOC
COLD-HEADER				
A	4.5 K	3 bar	DN50	SS316L
A'	4.5 K	3 bar	DN15	SS316L
B	2 K	27 mbar	DN40	SS316L
B'	2 K	27 mbar	DN15	SS316L
THERMAL SHIELD				
D	40 K	14 bar	DN32	CU
D'	40 K	14 bar	DN10	CU
E	60 K	13 bar	DN32	CU
E'	60 K	13 bar	DN10	CU
WARM (WPS)				
W	4.5 K-300	6 bar	DN10	SS304
S	2-292 K	1.05 bar	DN15	SS304
U	292 K	14 bar	DN15	SS304
REFERENCE				
OS				

SIGNAL TYPES (0.25 mm)

- Pneumatic (dash + //)
- Electric (dotted)
- Hydraulic (dash-dot)

INSTRUMENTS / SCOPE

- Field instrument (white box tag)
- Panel / safety (line)
- TP scope diamond (edge = TP layer)

Car: B=Building, C=Chemical, E=Electrical, G=Compressed gases
H=HVAC, L=Liquid waste, S=Solid waste, W=Water

CONSULTANT Mott MacDonald Bristol, UK	CLIENT SCK CEN Boeing 200, 2400 Mbl, BE MYRRHA / MINERVA Phase 1	REVISIONS Rev C1 B1 A1	Date 2026-06 2026-05 2026-04	Description v3 refinement - layers/legals v2 split sheets First issue	App'd ACR ACR ACR	APPROVALS Designed QSYS Drawn ACR Checked ACR Approved PM	MINERVA CryoCell - SCK CEN QCELL / LB - Cryogenic Circuits (40 K / 4.5 K / 2 K) DWG No: -NA.P501.PF8712 MMD Proj: 411966 Scale: NTS Suitability: S2 - FOR ACCEPTANCE Security: RESTRICTED Size / Std: A3 / ISO1628 / ISA-S.1 Sheet: Sheet of 2 - Cryogenic Variant: CONTROL-CENTRIC
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